

arc-agi

Taxonomía Deep Learning con Benchmarks ARC-AGI

Deep Learning	
└─ 1. Modelos Perceptivos	
└─ CNNs	
└─ Vision Transformers (ViT)	
└─ Multimodal Encoders	
└─ 2. Modelos Generativos	
└─ GANs	
└─ VAEs	
└─ Diffusion Models	
└─ Flow-based Models	
└─ 3. Modelos Secuenciales	
└─ RNN / LSTM / GRU	
└─ Transformers	
└─ State Space Models (Mamba, S4)	
└─ 4. Foundation Models (Base LLMs)	
└─ LLMs	
└─ GPT-3	ARC-AGI-1: ~0%
└─ GPT-4	ARC-AGI-1: ~0%
└─ GPT-4o	ARC-AGI-1: 5%
└─ GPT-4.1-Nano	ARC-AGI-1: 0% \$0.002/task
└─ Llama 4 Maverick	ARC-AGI-1: 4.4% \$0.0078/task ARC-AGI-2: 0%
└─ Grok 3	ARC-AGI-1: 5.5% \$0.0093/task ARC-AGI-2: 0%
└─ Claude 3.5 Sonnet	ARC-AGI-1: 14% (base, sin técnicas adicionales)
└─ Vision Foundation Models	
└─ Multimodal Foundation Models	
└─ 5. Neuro-Symbolic & Reasoning AI	
└─ Neuro-symbolic models	
└─ Program Induction Models	
└─ Test-Time Training (MIT/Cornell)	ARC-AGI-1: 47.5%
└─ CompressARC (76K params)	ARC-AGI-1: 20-34% ARC-AGI-2: 4% (sin pretraining)
└─ Neural Theorem Provers	
└─ Differentiable Reasoning Systems	
└─ 6. World Models & Cognitive Models	
└─ World Models	
└─ Predictive Processing Models	
└─ Active Inference Models	
└─ Cognitive Architectures	
└─ 7. **Reasoning-Centric Models (LRMs - Large Reasoning Models)**	
└─ TRM (Token Reasoning Models) - Single Chain-of-Thought	
└─ OpenAI o1 (low)	ARC-AGI-1: 20.5% \$0.43/task
└─ OpenAI o1 (medium)	ARC-AGI-1: 31% \$0.79/task
└─ OpenAI o1 (high)	ARC-AGI-1: 35% \$1.31/task
└─ DeepSeek R1-Zero	ARC-AGI-1: 14% \$0.11/task (sin SFT, solo RL)
└─ DeepSeek R1	ARC-AGI-1: 15.8% \$0.06/task ARC-AGI-2: 1.3%
└─ DeepSeek R1 (5/28/25)	ARC-AGI-1: 21.2% \$0.046/task (actualizado)
└─ OpenAI o3 (low)	ARC-AGI-1: 41% \$1.22/task ARC-AGI-2: 1.9%
└─ OpenAI o3 (medium)	ARC-AGI-1: 53% \$2.52/task ARC-AGI-2: 2.9%
└─ OpenAI o3 (high)	ARC-AGI-1: 60.8% \$0.50/task ARC-AGI-2: 6.5%
└─ OpenAI o3-pro (high)	ARC-AGI-1: 59.3% \$4.16/task ARC-AGI-2: 4.9%
└─ OpenAI o4-mini (low)	ARC-AGI-1: 21% \$0.05/task ARC-AGI-2: 1.6%
└─ OpenAI o4-mini (medium)	ARC-AGI-1: 42% \$0.23/task ARC-AGI-2: 2.3%
└─ OpenAI o4-mini (high)	ARC-AGI-1: 58.7% \$0.406/task ARC-AGI-2: 6.1%
└─ Grok 3 Mini (low)	ARC-AGI-1: 16.5% \$0.0099/task ARC-AGI-2: 0.4%
└─ Claude Sonnet 4 (16k)	ARC-AGI-1: 40% \$0.366/task ARC-AGI-2: 5.9%
└─ Claude Opus 4 (16k)	ARC-AGI-1: 35.7% \$1.25/task ARC-AGI-2: 8.6%
└─ Claude Opus 4.5 (64k)	ARC-AGI-2: 37.6% \$2.20/task SOTA comercial verificado
└─ Gemini 2.5 Flash	ARC-AGI-1: 33.3% \$0.037/task (mejor balance)
└─ Gemini 2.5 Pro	ARC-AGI-1: 33% \$0.569/task ARC-AGI-2: 3.8%
└─ Gemini 3 Pro	ARC-AGI-2: 31% \$0.81/task (baseline)
└─ Search-Augmented Reasoning Models (Test-Time Compute Scaling)	
└─ OpenAI o3-preview (low)	ARC-AGI-1: 75.7% \$200/task 33.5M tokens

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| | | └─ OpenAI o3-preview (high) ─ ARC-AGI-1: 87.5% | $4,560/task | 5.7B tokens (172x compute)
| | | └─ Claude 3.5 Sonnet + Evolutionary TTC ─ ARC-AGI-1: 53.6% (Jeremy Berman)
|
| └─ **Tiny Recursive Models (TRM) - Redes Pequeñas con Razonamiento Recursivo** 
| | └─ TRM (7M params) ───────── ARC-AGI-1: 45% | ARC-AGI-2: 8% | Paper Award 1st 2025
| | (https://arxiv.org/abs/2510.04871)
| | | └─ Solo 2 capas, <0.01% parámetros de LLMs, supera DeepSeek R1, o3-mini, Gemini 2.5 Pro
| | | └─ HRM (27M params) ───────── ARC-AGI-1: ~40% | Hierarchical Reasoning Model (precursor)
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| └─ HRM (Hierarchical Reasoning Models)
| | └─ Hierarchical planners
| | └─ Modular reasoning networks
| | └─ Multi-level cognitive models
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| └─ 8. Agentic AI Systems
| | └─ Autonomous Agents
| | └─ Tool-using Models
| | | └─ SLM Fine-tuned (OPT-350M) ─ ToolBench: 77.55% (supera ChatGPT-CoT 26%)
| | └─ Planning Agents
| | └─ Multi-agent Systems
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| └─ 9. **Soluciones Híbridas / Especializadas (Kaggle ARC Prize)**
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| | └─ ARC Prize 2024 (ARC-AGI-1)
| | | └─ the ARChitects (1°) ─── 53.5% (Program synthesis + search) (https://da-fr.github.io/arc-
| | | prize-2024/the_architects.pdf) usando x4 NVIDIA T4
| | | └─ Guillermo Barbadillo (2°) ─ 40%
| | | └─ alijs (3°) ───────── 40%
| | | └─ William Wu (4°) ───────── 37%
| | | └─ Ensemble Solutions ─── ~81% (combinación de múltiples soluciones)
| |
| | └─ ARC Prize 2025 (ARC-AGI-2) (https://arxiv.org/pdf/2601.10904)
| | | └─ NVARC (1°) ───────── 24.03% | $0.20/task | SOTA Kaggle
| | | (https://drive.google.com/file/d/1vkEluaaJTzaZiJL69TkZovJUkPSDH5Xc/view)
| | | └─ the ARChitects (2°) ─── 16.53% | 2D-aware masked-diffusion LLM
| | | (https://lambdalabsml.github.io/ARC2025_Solution_by_the_ARChitects/)
| | | └─ MindsAI (3°) ───────── 12.64% | Test-time training pipeline (https://arxiv.org/abs/2506.14276)
| | | └─ Lonnie (4°) ───────── 6.67%
| | | (https://www.kaggle.com/competitions/arc-prize-2025/writeups/arc-prize-2025-competition-writeup-5th-place)
| | | └─ G. Barbadillo (5°) ─── 6.53%
| | | (https://ironbar.github.io/arc25/05_Solution_Summary/#vision-search-and-learn)
|
| └─ 10. **Model Refinement Solutions (Refinement Loops)**
| | └─ Poetiq (Gemini 3 Pro Ref) ─ ARC-AGI-2: 54% | $31/task | SOTA refinement
| | └─ Poetiq (Claude Opus 4.5 Ref) ─ ARC-AGI-2: ~54% | $60/task
| | └─ SOAR (Self-Improving LLM) ─ ARC-AGI-1: 52% | Paper Award 2nd 2025
| | └─ Evolutionary Program Synthesis (E. Pang) ─ Paper Award Runner-up
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Resumen de Benchmarks ARC-AGI (Actualizado Dic 2025)

ARC-AGI-1 Semi-Private Eval (Ranking por rendimiento)

Modelo/Sistema	Score	Costo/Tarea	Notas
o3-preview (high)	87.5%	\$4,560	172x compute, 5.7B tokens
Ensemble Kaggle	~81%	-	Combinación de soluciones
o3-preview (low)	75.7%	\$200	Primer puesto leaderboard 2024
o3 (high)	60.8%	\$0.50	-
o3-pro (high)	59.3%	\$4.16	-
o4-mini (high)	58.7%	\$0.406	-
Claude 3.5 + EvoTTC	53.6%	-	Evolutionary Test-Time Compute
the ARChitects (2024)	53.5%	-	Ganador Kaggle ARC Prize 2024
o3 (medium)	53%	\$2.52	-
SOAR	52%	-	Self-Improving Evolutionary Program Synthesis
TTT MIT/Cornell	47.5%	-	Test-Time Training
TRM (7M params)	45%	-	 Tiny Recursive Model - Paper Award 1st
o3 (low)	41%	\$1.22	-

Modelo/Sistema	Score	Costo/Tarea	Notas
Claude Sonnet 4 (16k)	40%	\$0.366	-
o4-mini (medium)	42%	\$0.23	-
Claude Opus 4 (16k)	35.7%	\$1.25	-
o1 (high)	35%	\$1.31	-
Gemini 2.5 Flash	33.3%	\$0.037	Mejor balance costo/rendimiento
Gemini 2.5 Pro	33%	\$0.569	-
o1 (medium)	31%	\$0.79	-
DeepSeek R1 (5/28)	21.2%	\$0.046	Actualizado
o4-mini (low)	21%	\$0.05	-
o1 (low)	20.5%	\$0.43	-
CompressARC (76K)	20-34%	-	Sin pretraining, MDL-based
Grok 3 Mini (low)	16.5%	\$0.0099	-
DeepSeek R1	15.8%	\$0.06	Open source
DeepSeek R1-Zero	14%	\$0.11	Sin SFT, solo RL
Claude 3.5 Sonnet	14%	-	Base, sin técnicas adicionales
Grok 3	5.5%	\$0.0093	-
GPT-4o	5%	-	Base LLM
Llama 4 Maverick	4.4%	\$0.0078	-
GPT-4	~0%	-	Base LLM
GPT-3	0%	-	Base LLM
Humanos	~85%	~\$5	Referencia humana

ARC-AGI-2 (lanzado Marzo 2025)

Modelo/Sistema	Score	Costo/Tarea	Notas
Poetiq (Gemini 3 Pro Ref)	54%	\$31	 SOTA refinement solution
Claude Opus 4.5 (64k)	37.6%	\$2.20	 SOTA comercial verificado
Gemini 3 Pro	31%	\$0.81	Baseline
NVARC (Kaggle 1st)	24.03%	\$0.20	 SOTA Kaggle 2025
the ARCHitects	16.53%	-	2D-aware masked-diffusion
MindsAI	12.64%	-	Test-time training
Claude Opus 4 (16k)	8.6%	\$1.93	-
TRM (7M params)	8%	-	 Tiny Recursive Model
o3 (high)	6.5%	\$0.834	-
o4-mini (high)	6.1%	\$0.856	-
Claude Sonnet 4 (16k)	5.9%	\$0.486	-
o3-pro (high)	4.9%	\$7.55	-
CompressARC (76K)	4%	-	Sin pretraining
Gemini 2.5 Pro	3.8%	\$0.813	-
o3 (medium)	2.9%	-	-
o4-mini (medium)	2.3%	-	-
o3 (low)	1.9%	-	-
o4-mini (low)	1.6%	-	-
DeepSeek R1	1.3%	\$0.08	-
Grok 3 Mini (low)	0.4%	\$0.013	-

Modelo/Sistema	Score	Costo/Tarea	Notas
Grok 3	0%	\$0.14	-
Llama 4 Maverick	0%	\$0.012	-
Humanos	>95%	-	Sin entrenamiento previo

Insights Clave (Actualizado 2025)

- Los LLMs base fracasan** en ARC-AGI porque dependen de memorización, no razonamiento adaptativo
 - Los Reasoning Models (o1, R1, o3)** mejoran mediante Chain-of-Thought, pero tienen límites
 - El test-time compute scaling** permite mejoras logarítmicas pero con rendimientos decrecientes
 - Tiny Recursive Models (TRM)** logran 45% en ARC-AGI-1 con solo 7M parámetros (<0.01% de LLMs)
 - ARC-AGI-2 sigue sin resolver:** incluso los mejores modelos <10%, humanos >95%
 - Refinement Loops** son el tema central de 2025 - mejoran performance iterativamente
 - La eficiencia importa:** Gemini 2.5 Flash ofrece el mejor ratio costo/rendimiento para ARC-AGI-1
 - No hay ganador claro:** cada modelo tiene trade-offs entre accuracy, costo y velocidad
 - ARC-AGI-3** (2026) introducirá razonamiento interactivo: exploración, planificación, memoria

Bibliografía Relevante

Papers con resultados ARC-AGI

- "Less is More: Recursive Reasoning with Tiny Networks"** (Oct 2025)
 - Autor: Alexia Jolicoeur-Martineau
 - arXiv: 2510.04871
 - Resultado: TRM (7M params) → ARC-AGI-1: 45%, ARC-AGI-2: 8%
 - 🏆 Paper Award 1st Place - ARC Prize 2025
 - "Small Language Models for Efficient Agentic Tool Calling"** (Dec 2025)
 - Autores: Jhandi, Kazi, Subramanian, Sendas
 - arXiv: 2512.15943
 - Resultado: OPT-350M fine-tuned → ToolBench: 77.55% (vs ChatGPT-CoT: 26%)
 - "On the Measure of Intelligence"** (2019)
 - Autor: François Chollet
 - arXiv: 1911.01547
 - Introduce ARC-AGI benchmark
 - "The Surprising Effectiveness of Test-Time Training for Abstract Reasoning"** (2024)
 - MIT/Cornell
 - Resultado: ARC-AGI-1: 47.5%

Enlaces imprescindibles

<https://lewish.io/posts/arc-agi-2025-research-review#ttt-ttft>

<https://lewish.io/posts/how-to-beat-arc-agi-2>